Quantum Checkers Administrator Guide

A practical guide for managing classes, students and puzzles. No coding or database tools are needed.

Game: https://quantum-checker.onrender.com/
Admin dashboard: https://quantum-checker.onrender.com/admin/

1. Sign in

1. Open the admin dashboard.
2. Enter the administrator **username** and password supplied to you. If the account uses an email address as its username, enter the whole address.
3. Click **Log in**. You should see sections such as **Classrooms**, **Student profiles** and **Game levels**.

> Your administrator password is supplied separately. This guide contains no passwords. The free website can take a little time to wake up after being idle; wait and retry if the first page loads slowly.

2. Add a classroom before students register

1. In the dashboard, click **Add** next to **Classrooms**.
2. Enter a **Classroom name**, for example **Fall 2026**.
3. Click **Save**.
4. Open the game's registration page and check the **Classroom (optional)** dropdown. Your new classroom should appear. Refresh the page if it was already open.
5. Give students the game link and the exact classroom name to select.

Students select their classroom when registering. Classroom selection is currently optional, so remind your students to select **Fall 2026** rather than leave **Unassigned**. Students who register without a class can be assigned one later in **Student profiles**.

To rename a classroom, open **Classrooms**, click its name, edit it and save. Existing students stay assigned to that classroom. Deleting a classroom leaves its students unassigned; it does not delete their accounts.

3. Understand student accounts

Field	Meaning
Username	The name the student uses to log in. It is separate from their student ID.

Field	Meaning
Student ID	Must start with uppercase **D** followed by digits only, for example `D00123456`. Each ID can belong to only one account across all classrooms. Zeros after D are preserved.
Classroom	The student's class, for example **Fall 2026**.
Email	Optional contact information. An email address is not automatically the login username.

Examples of invalid student IDs: `d12345`, `12345`, `D12A45`, `D12-45` and `D 12345`. Administrators and existing accounts can have a blank student ID; new registrations must supply a valid one.

4. View and correct student details

1. Open **Student profiles**.
2. Use the search box to find a username or student ID. Use the **Classroom** filter to see one class.
3. Click the student's username.
4. Correct the **Student ID** or choose a different **Classroom**.
5. Click **Save**. A duplicate or incorrectly formatted ID produces an error; correct it and save again.

Profiles are created automatically when accounts are created. You do not need to add profiles yourself. Students cannot currently edit their ID or classroom themselves; ask them to contact you for corrections.

Reset a forgotten password

A full administrator can open **Users**, click the student's username and follow the password-change link in the password section. Set a new password, save it and give it to the student privately. Students then log in with their existing username. There is no automatic email password-reset service.

5. Edit a game level

1. Open **Game levels** and click a level.
2. Change its **Title** and **Instructions**. These appear in the game.
3. Use the checked **Gate x**, **Gate z**, **Gate h** and **Gate cz** boxes to choose which moves students can use.
4. Set the target circles using the table below. Keep **Advanced win conditions** as `[]` for ordinary puzzles.
5. Click **Save**. If the target cannot be reached with the allowed gates, the page shows an error and does not save the change.
6. Open the game, choose the level and test it. Refresh a level that was already open to load your changes.

Target value	What students see
`1`	Black circle
`0`	White circle
`0.5`	Outlined circle
Blank	Dashed circle: any value is accepted

The circle labels identify the left qubit, right qubit or a correlation between them. Their row and column are shown in brackets. Keep target values to `0`, `0.5` and `1` for these puzzles.

Set the starting board

Initial state describes moves made before the student begins. Leave it blank for the default starting board. If needed, enter operations separated by commas:

Example	Meaning
`h:0`	Apply H to the left qubit.
`x:1`	Apply X to the right qubit.
`h:0, h:1`	Apply H to both qubits, one after the other.
`cz`	Apply CZ to both qubits together.

Use lowercase `x`, `z`, `h` or `cz`. For X, Z and H, `0` means left and `1` means right. CZ takes no qubit number. Initial-state operations are separate from the gates students are allowed to use.

Add a simple new puzzle

1. Click **Add** next to **Game levels**.
2. Choose a positive **Level** number that is not already used, such as **5** if levels 1–4 are present.
3. Set **Title** to “Flip the left qubit” and **Instructions** to “Select the left qubit and apply X.”
4. Leave **Initial state** blank. Check only **Gate x**.
5. Set **Left qubit · Z (1,1)** to `0`. Leave all other target circles blank.
6. Leave **Win conditions** as `[]` and click **Save**.
7. Play the new level. Select **Left qubit** and apply X once; it should show **Puzzle solved!**

> Changing a level's starting state, targets or allowed gates restarts existing attempts when that level is reopened. Deleting a level also removes its saved game sessions. Keep level numbers stable once students start playing.

6. Review game sessions

Open **Game sessions** to see attempts by student and level. Use the **Is win** and **Game level** filters to narrow the list. Opening a session shows its current moves and state; these details are read-only. A student's session is updated as they play and can be reset, so this list is not a permanent grading history.

7. Give another person management access

A full administrator can create a separate management account without sharing their own password:

1. Open **Users**, click **Add**, enter a username, set a password and save.
2. Open that user again. Ensure **Active** and **Staff status** are checked.
3. In **Groups**, move **Level editors** into the selected groups and save.
4. Give the person the admin-dashboard link, their username and password privately.

The **Level editors** group can manage levels and classrooms, view and correct student profiles, and view game sessions. It cannot manage user passwords or grant account permissions. Leave **Superuser status** unchecked for this role. If someone must take over all account administration, a full administrator can instead grant **Superuser status**, which gives access to all admin functions.

8. Common problems

Problem	What to do
A class is missing at registration.	Check that you saved it in Classrooms, then ask the student to refresh the registration page.
An ID is already in use.	Search Student profiles for that ID. Help the student use their existing account rather than create another one.
A level will not save.	Read the error. Check the target values and allowed gates, or start with the simple puzzle above.
The admin page denies access.	Ask a full administrator to check the account's Active, Staff status and group permissions.

Problem	What to do
A student is in the wrong class.	Correct Classroom in their Student profile and save.
